WORKED SOLUTION

STEP I 1989, Question 10

Mechanics • Velocity-Dependent Resistance

> **STEP I 1989, Question 10**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/89-s1-q10

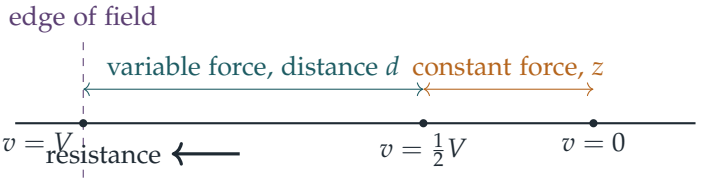
The key idea

The resistance depends on the speed v, not on time, and we want a *distance*. That is the signal to write the acceleration as $v\,\dfrac{\mathrm{d}v}{\mathrm{d}x}$, which turns Newton's second law into a separable equation linking v and x directly. The second part is a constant-deceleration problem, so the only work is to find the force at the moment of the change and let it vary by $\pm 10\%$.

Solution

The model

We treat the spaceship as a particle of mass M moving along a straight line. Let x be the distance travelled since entering the field, and v the speed, with $v = V$ at $x = 0$. The only force is the resistance, which acts backwards, so the speed decreases throughout.



Distance to halve the speed

Taking the direction of motion as positive and writing the acceleration as $v\,\dfrac{\mathrm{d}v}{\mathrm{d}x}$, Newton's second law gives

$$Mv\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{M\omega\,(v^2 + V^2)}{v} \qquad \Longrightarrow \qquad \frac{v^2}{v^2 + V^2}\,\mathrm{d}v = -\omega\,\mathrm{d}x.$$

Ask yourself How do we integrate $\dfrac{v^2}{v^2+V^2}$?

Write the numerator as $(v^2+V^2)-V^2$, so that

$$\frac{v^2}{v^2+V^2} = 1 - \frac{V^2}{v^2+V^2}, \qquad \int \frac{v^2}{v^2+V^2}\,\mathrm{d}v = v - V\arctan\frac{v}{V} + C,$$

using the standard integral $\int \frac{\mathrm{d}v}{v^2+V^2} = \frac{1}{V}\arctan\frac{v}{V}$.

Let d be the distance travelled when the speed has fallen to $\frac{1}{2}V$. Integrating from $(x,v)=(0,V)$ to $(d,\frac{1}{2}V)$,

$$\begin{aligned}\omega d &= \int_{V/2}^{V} \frac{v^2}{v^2+V^2}\,\mathrm{d}v = \Big[v - V\arctan\frac{v}{V}\Big]_{V/2}^{V} \\ &= \frac{V}{2} - V\Big(\arctan 1 - \arctan\tfrac{1}{2}\Big) = V\Big(\frac{1}{2} - \frac{\pi}{4} + \arctan\frac{1}{2}\Big).\end{aligned}$$

So the required distance is

$$d = \frac{V}{\omega}\Big(\frac{1}{2} - \frac{\pi}{4} + \arctan\frac{1}{2}\Big) = \frac{V}{\omega}\Big(\frac{1}{2} - \arctan\frac{1}{3}\Big) \approx 0.178\,\frac{V}{\omega}.$$

The second form uses $\arctan 1 - \arctan\frac{1}{2} = \arctan\dfrac{1-\frac{1}{2}}{1+\frac{1}{2}} = \arctan\frac{1}{3}$, from the tangent subtraction formula (both angles lie in $(0,\frac{\pi}{2})$, so there is no ambiguity). Either form is a complete answer.

The constant force

Immediately before the change, $v=\frac{1}{2}V$, so the resistive force has magnitude

$$\frac{M\omega(\frac{1}{4}V^2+V^2)}{\frac{1}{2}V} = \frac{5}{2}M\omega V.$$

The new constant force F is within 10% of this, so

$$\frac{9}{10}\cdot\frac{5}{2}M\omega V \le F \le \frac{11}{10}\cdot\frac{5}{2}M\omega V, \qquad \text{i.e.} \qquad \frac{9}{4}M\omega V \le F \le \frac{11}{4}M\omega V.$$

With a constant force the deceleration $a = F/M$ is constant, so $v^2 = u^2 - 2as$ with $u=\frac{1}{2}V$ and final speed 0 gives

$$z = \frac{(V/2)^2}{2a} = \frac{MV^2}{8F}.$$

This decreases as F increases, so the largest force gives the smallest z and vice versa:

$$\frac{MV^2}{8\cdot\frac{11}{4}M\omega V} \le z \le \frac{MV^2}{8\cdot\frac{9}{4}M\omega V}, \qquad \text{that is} \qquad \boxed{\frac{V}{22\,\omega} \le z \le \frac{V}{18\,\omega}}$$

(with strict inequalities if "within 10%" is read strictly). If the force were exactly unchanged, $z = V/(20\omega)$, in the middle of this range.

Checks

- Dimensions: ω has units s^{-1} (since $M\omega v$ must be a force), so V/ω is a length, as both answers require.
- The coefficient $\frac{1}{2} - \arctan\frac{1}{3} \approx 0.1782$ is positive and less than $\frac{1}{2}$. It should be: the ship travels at speeds between $\frac{1}{2}V$ and V, and the deceleration is at least $2\omega V$, so the trip lasts at most $1/(4\omega)$ and covers less than $V/(4\omega)$.
- Stepping $\dot{v} = -\omega(v^2 + V^2)/v$ numerically with $V = \omega = 1$ gives 0.17825 for the distance at which $v = \frac{1}{2}$, and stepping the constant-force phase gives $z = 0.0455$, 0.0500, 0.0556 for forces of 110%, 100% and 90%, matching $\frac{1}{22}$, $\frac{1}{20}$, $\frac{1}{18}$.

Common slips

- Using $\dfrac{\mathrm{d}v}{\mathrm{d}t}$ instead of $v\dfrac{\mathrm{d}v}{\mathrm{d}x}$: this gives the time, not the distance, and an extra integration is then needed.
- Pairing the bounds the wrong way round: a *larger* force gives a *shorter* stopping distance, so 1.1 times the force goes with the lower limit $V/(22\omega)$.

Beyond STEP

Writing $a = v\,\mathrm{d}v/\mathrm{d}x$ is the first glimpse of a bigger idea: for a one-dimensional force depending only on position or speed, one can trade time for another variable and reduce the order of the equation, and when the force depends only on position this leads to energy conservation and the phase-plane picture of motion. The second part is a small exercise in *sensitivity*: how an uncertainty in the data (here $\pm 10\%$ in the force) propagates to the answer (roughly $\mp 10\%$ in z). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats velocity-dependent forces in this spirit.

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